BTP Progress Update — after mid-semester

Kaveri Neeli (EP230051010) | A Multi-Agent System for Time-Series Modelling across Scientific and Financial Domains | 7 October 2026

1. Re-checked whether the earlier results held up

The query labels used before mid-sem were stale — they expected a forecasting model we had already retired. After correcting them and re-running all three selection strategies on 141 labelled queries:

Strategy	Accuracy	Correct	Time taken
filter_only (no LLM)	95.0%	134 / 141	0.0 s
hybrid (LLM on shortlist)	80.8%	114 / 141	53 min
llm_only (unconstrained)	65.2%	92 / 141	63 min

Earlier the LLM looked nearly as good as the deterministic filter. It was not — the old labels were flattering it. The capability constraint does almost all the useful work, and costs nothing per query.

2. Found and fixed a routing gap

The stock risk model was implemented but never registered as a task, so the selector could never reach it — the chatbot only got there via a separate keyword shortcut. After registering it, all ten risk queries route correctly and none of the existing 123 changed.

3. Added a proper significance test to the anomaly results

Anomaly-detection windows overlap heavily (18 of 24 points on Kepler), so treating them as independent understates the variance. Resampling whole targets instead, **neither headline margin is distinguishable from zero**: the naive test reported p = 1.0000 where the correct interval spans zero. This is reported rather than the flattering version.

4. Scored the LSTM that had been sitting unused

It had never been included in the benchmark, so no performance claim was backed by a number. Held-out PR-AUC **0.2591**, against **0.2595** for moving variance — a one-line statistic — and it costs 909 s against 3.5 s. The hypothesis that the order of points within a window carries signal is not supported.

5. Changed direction on the finance side

Every price model loses to “tomorrow equals today”, which is the textbook random-walk result rather than a modelling failure. Confirmed on our own data: daily returns have autocorrelation **+0.010**, essentially zero, while volatility has **+0.459**. So the target was changed rather than the model.

Forecasting realized volatility instead, the HAR model cuts forecast loss **47%** against the naive baseline across 59,709 out-of-sample forecasts, and naive is the best model on **0 of 48 stocks**. This is the first result in the project that clearly beats its baseline — which matters as a positive control: a framework that only ever reports “nothing works” cannot be told apart from a broken instrument.

6. Working on now: widening the volatility comparison

Ten models under one protocol, arbitrated by a Model Confidence Set — the standard test that returns the set of models which cannot be statistically separated from the best.

Model	What it adds	Confidence set
naive	tomorrow = today (the null)	eliminated
HAR	Heterogeneous Autoregressive: daily / weekly / monthly averages	kept
Log-HAR	the same regression fitted in logs	kept *
HAR-L	leverage: a fall raises volatility more than a rise	kept
HAR-IV	India VIX, the market's implied volatility (free from NSE)	kept
HAR-OVN	the overnight gap a daily range estimator cannot see	kept
Pooled HAR	one model across all 50 stocks instead of 50 separate	eliminated
Linear AR(22)	HAR with its constraint removed, 22 free parameters	eliminated *
GBM	gradient boosting, to test whether non-linearity helps	eliminated
Chronos-Bolt	Amazon's pretrained foundation model, zero-shot	eliminated *

* Scored in earlier comparisons against HAR rather than inside this ten-model set. A single joint confidence set covering all ten, plus GARCH and CARR, is the next step — the confidence set is a joint test, so rows from different runs are not strictly comparable.

Still to add: GARCH(1,1) and GJR-GARCH, the classical volatility benchmarks, and CARR, which is built specifically for daily high-low range data of the kind we use. Also running a sweep over the fitting scheme (training window length and refit frequency), since recent work argues this matters more than the choice of model — something never tested in our own code.

The theme that keeps recurring

In three separate places the simple option beats the complex one, and honest measurement is what reveals it: a deterministic filter beats an unconstrained language model, a one-line statistic beats an LSTM, and a 3-parameter regression beats a 22-parameter one and a pretrained transformer.

All figures above are read from committed result files in the repository. HAR, GARCH, conformal prediction and the Model Confidence Set are implemented from their published definitions and cited; the contribution is the selection framework and the uniform measurement protocol applied across anomaly detection, price and volatility.